

GAHAN SARAIYA

☎ +91 7359508173 ◊ ✉ gahansaraiya@gmail.com ◊ in [linkedin.com/in/gahan-saraiya](https://www.linkedin.com/in/gahan-saraiya) ◊ 🐙 github.com/gahan9

PROFESSIONAL SUMMARY

Senior DevOps and MLOps Engineer with 8+ years of experience designing, implementing, and managing scalable cloud infrastructure, CI/CD pipelines, and machine learning operations. Proven expertise in containerization, Infrastructure as Code (IaC), multi-cloud platforms (AWS, Azure, GCP), Kubernetes orchestration, and automated ML model deployment pipelines. Strong background in Python automation, Docker containerization, Terraform, Jenkins/GitHub Actions CI/CD, and observability solutions. Experienced in managing AI/ML workload execution on HPC devices, implementing MLOps practices for model training, deployment, and monitoring at scale.

TECHNICAL SKILLS

Programming Languages	Python, Bash Shell Scripting, SQL, YAML, JSON, PowerShell
DevOps & CI/CD Tools	Jenkins, GitHub Actions, GitLab CI/CD, Terraform, Ansible, Docker, Docker Compose, Kubernetes (K8s), Helm, Git, TeamCity, JFrog Artifactory,
Cloud Platforms	AWS (EC2, S3, CloudWatch, VPC, Lambda, CloudFront, ECR, ECS, EKS, SageMaker, IAM, Route53), Azure (DevOps, OpenAI, Container Registry), GCP (Cloud Storage, Cloud Functions), Digital Ocean
MLOps & ML Frameworks	PyTorch, TensorFlow, LangGraph, FAISS, MLflow, Model Serving, Model Deployment, Model Monitoring, AWS SageMaker, MLOps Pipelines, Model Versioning
Infrastructure & Monitoring	Infrastructure as Code (IaC), Container Orchestration, Cloud Architecture, System Monitoring, Logging, Observability, Performance Optimization, Auto-scaling, Load Balancing
APIs & Services	RESTful APIs, Microservices Architecture, API Gateway, Service Mesh, Redfish API, Selenium Automation

PROFESSIONAL EXPERIENCE

Senior System Software Designer at AMD India Pvt Ltd	Mar 2024 - Present
- Architected and managed distributed workload execution pools across multiple geographic regions, implementing CI/CD pipelines on Jenkins for automated build, test, and deployment workflows - Led release management operations for ROCm builds across production, staging, and development environments, managing complex matrix configurations of operating systems and toolchains - Designed and implemented Docker containerization strategies with enterprise container registry (JFrog Artifactory), optimizing image builds and deployment pipelines - Migrated platform provisioning infrastructure to Infrastructure as Code (IaC) using Terraform, enabling automated infrastructure deployment and reducing manual configuration errors by 70% - Developed and maintained MLOps pipelines for AI/ML workload execution on HPC devices (GPUs), integrating automated testing, model versioning, and deployment workflows - Implemented Kubernetes-based container orchestration solutions for managing GPU workloads, optimizing resource allocation and ensuring high availability - Created automated monitoring and alerting systems for infrastructure health, performance metrics, and deployment status using CloudWatch and custom observability tools - Conducted functional and performance benchmarking of HPC devices (GPUs) by developing and executing comprehensive test suites aligned with industry standards	
Senior Firmware Development Engineer at Intel Technology India Pvt Ltd	Oct 2023 - Mar 2024

- Designed and implemented platform provisioning and infrastructure automation for large-scale HPC product deployments, reducing manual setup time by 70%
- Architected Platform Orchestration Layer for Universal Scalable Firmware (USF), enabling automated configuration management and deployment workflows
- Built complete Infrastructure as Code (IaC) solution using Terraform for on-demand development environments on Kubernetes clusters, including node provisioning from scratch
- Implemented Kubernetes-based container orchestration for development environments, enabling rapid scaling and resource optimization with automated node management, pod scheduling, and resource allocation policies
- Integrated Azure OpenAI services for automated code review, security vulnerability scanning, and code quality assessment, improving development productivity
- Integrated GitHub Actions and JIRA for end-to-end CI/CD pipeline management, implementing continuous deployment and automated testing through CloudBees Jenkins
- Managed hybrid Jenkins infrastructure with bare-metal and Docker containerized nodes, optimizing build performance and resource utilization

Firmware Development Engineer at Intel Technology India Pvt Ltd

Jun 2020 - Sept 2023

- Led next-generation client platform architecture design at USF POL, owning domains for Automation Framework, Platform Provisioning, Applied Generative AI, and Quantum Firmware Design
- Managed end-to-end Software Development Life Cycle (SDLC), including licensing, legal compliance, IP planning, code security, and static code analysis integration
- Configured and maintained Jenkins CI/CD pipelines with security scanning tools (Snyk, Coverity), implementing automated security vulnerability detection in build processes
- Enhanced GitHub code review automation with Python packaging validation, linting (Bandit), and Flask-based Jenkins pipeline integration for improved code quality
- Implemented automated testing frameworks for firmware functional validation using TeamCity, reducing manual testing effort by 60%
- Built and maintained continuous integration pipelines for automated build, test, and deployment workflows
- Established DevSecOps practices by integrating security scanning, code analysis, and compliance checks into CI/CD pipelines
- Mentored team members on DevOps best practices, automation frameworks, and CI/CD pipeline development

Software Developer at Quixom Technology

Jul 2017 - Apr 2018

- Developed SaaS-based employee management system with RESTful API architecture, implementing microservices design patterns
- Built real-time data processing pipelines using Python socket programming for continuous IoT device data ingestion and analytics
- Designed and deployed cross-platform streaming service with Django REST Framework (DRF) backend and web portal frontend
- Implemented automated data processing workflows for handling large-scale data streams from IoT devices

Freelance Full-stack Developer

2015 - Apr 2019

- Designed and deployed cloud-based data gathering and web scraping solutions on Google Cloud Platform (GCP) using Selenium automation

- Built scalable data processing pipelines for automated data collection, storage, and analysis
- Developed inventory management system using Django framework with automated workflows for stock management and reporting
- Implemented cloud infrastructure solutions for high-availability web applications

KEY PROJECTS

Agentic AI for Software Engineering Modernization 2025 - Present

Designed and implemented end-to-end GenAI engineering workflows and automated repository management agents. Built custom tool integrations and Git hooks compatible with Claude and Cursor. Engineered robust security compliance layers, sandboxed action verification, and strict token spend/quota restriction mechanisms to ensure budget-compliant and secure execution on private corporate systems.

Automated Context-Aware Task Routing & Triage with LangGraph 2025 - Present

Developed and deployed automated, context-aware ticket triage and resolution pipelines. Built graph-based LLM routing logic with LangGraph, semantic classification using FAISS database indexes, and connectors to GitHub, Jira, and historical system data. Integrated Vertex AI, OpenAI, and local LLM endpoints behind a custom quality verification matrix for developer-facing advisory content.

DevOps Management for Cluster Validation 2025 - Present

Created highly consistent CI/CD infrastructure for validation of heterogeneous computing clusters. Designed secure S3-based log shipping mechanisms with custom IAM configuration, and configured fine-grained security policies and network validation checks complying with multi-region cloud regulations.

GPU Performance Analysis and MLOps Pipeline for Data Center Workloads 2024 - Present

Designed and implemented end-to-end MLOps pipelines for automated ML model training, deployment, and monitoring on GPU clusters. Built Docker-based containerization strategies and Terraform IaC solutions for automated GPU cluster provisioning. Engineered comprehensive monitoring solutions for GPU telemetry using System Management Interface (SMI). Integrated automated testing, model validation, and deployment workflows into Jenkins pipelines for continuous ML model delivery.

MLOps Pipeline for Generative AI Code Review Automation 2023 - 2024

Built automated MLOps pipelines integrating Azure OpenAI (GPT-3.5, GPT-4) for intelligent code analysis and review automation. Integrated AI-powered code review into Jenkins CI/CD pipelines, enabling automated code quality checks and security vulnerability detection. Enhanced security posture by automating code security scanning, vulnerability assessment, and compliance checking, reducing manual code review time by 50%.

Infrastructure Automation and Platform Provisioning 2023 - 2024

Developed Terraform-based IaC solutions for automated infrastructure provisioning and configuration management. Built end-to-end Jenkins pipelines for automated remote BMC and BIOS firmware flashing, reducing deployment time by 75%. Integrated JFrog Artifactory with Jenkins pipelines for automated OS image deployment and version management. Implemented automated system configuration using Redfish API for Baseboard Management Controller (BMC) operations.

DevOps Automation Framework for Firmware Development 2020 - Present

Designed and implemented fully automated system configuration framework, eliminating manual BIOS configuration intervention for thousands of devices. Established comprehensive DevSecOps practices including legal compliance automation, security scanning, licensing validation, and static code analysis in CI/CD pipelines. Built end-to-end continuous deployment and testing pipelines using Jenkins with tox configurations. Developed and maintained open-source automation tools: <https://github.com/intel/xml-cli>

EDUCATION

M.Tech Computer Science
Nirma University, GPA: 8.49

Jul 2018 - Jun 2020

CERTIFICATIONS

2025	Essential Terraform in AWS - LinkedIn Learning
2025	AMD ROCm Star - Application Developer - AMD Adaptive Computing Training Team
2025	On-Device AI - Qualcomm & DeepLearning.ai
2021	Product Assurance and Security Yellow Belt - Software - Intel
2020	Machine Learning Specialization (5 courses) - Coursera (DeepLearning.ai)

ACHIEVEMENTS & CONTRIBUTIONS

- Top 10 in AI Hackathon: Hack2Skill Prompt Wars by Google (In-person, Bangalore) (2026)
- Rank #3 in Virtual Prompt Wars Hackathon (2026)
- Presented Firmware Configuration at UEFI Developers Conference & Plugfest (2023)
- Tech Talk on Harnessing Multiprogramming across CPU, GPU and QPU (2023)
- Expert Lecture on CUDA Programming (2021)
- 4000+ reputation on Stack Overflow (2019-Present)
- Rank #1 for Python challenges on HackerRank (2018)
- Rank #112 from 200,000+ participants in Techgig's National Coding Contest (2018)

PUBLICATIONS & PRESENTATIONS

- [HIP Parallel Programming with ROCm](#) Lecture Series, YouTube (2025)
- [Beyond the Power — Data Center & BMC](#) Lecture Series, YouTube (2025)
- [System Design for Parallel Programming](#) Lecture Series, YouTube (2025)
- [Firmware Configuration – Past, Present, and Future](#), UEFI DevCon 2023